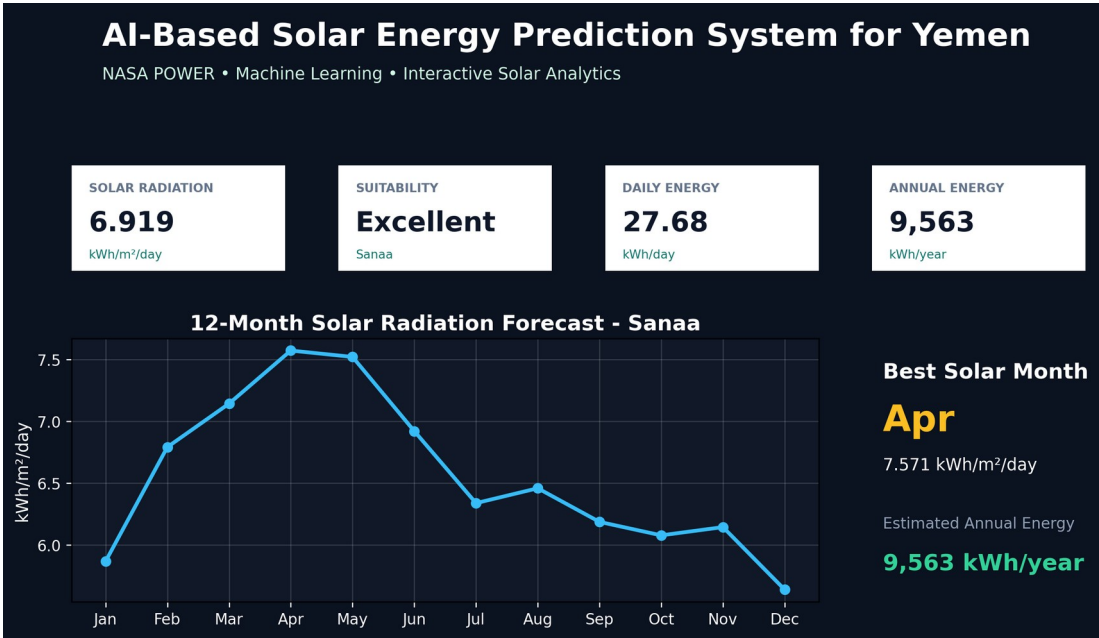


SANA'A UNIVERSITY

FACULTY OF ENGINEERING

AI-BASED SOLAR ENERGY
PREDICTION SYSTEM FOR YEMEN

Machine Learning Project Report



Professional Streamlit dashboard developed for interactive prediction, analytics and model testing.

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Academic Year	2025-2026
Data Source	NASA POWER (2015-2025)

Data → Pre-processing → EDA → Model Training → Evaluation → Streamlit Deployment

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Executive Summary

This project develops an artificial-intelligence-based system for predicting daily solar radiation and estimating photovoltaic energy potential across selected Yemeni cities. The workflow starts with programmatic acquisition of solar and meteorological observations from the NASA POWER API, continues through data-quality checking, feature engineering and exploratory analysis, and then compares three supervised regression algorithms. The best model is integrated into a Streamlit dashboard that supports interactive prediction, 12-month scenario forecasting, city comparison, model-performance review and a Yemen solar-potential map.

32,144 Records	8 Cities	2015-2025 Period	Random Forest Best Model
0.636 Best R ²	0.348 MAE	0.505 RMSE	Streamlit Deployment

Project Objectives

- Build a Yemen-focused solar-radiation dataset from an authoritative and reproducible public source.
- Validate the dataset and create useful temporal features before training.
- Visualize spatial, seasonal and meteorological relationships in the data.
- Train and compare multiple machine-learning regression algorithms using the same split and metrics.
- Select and save the best-performing pipeline for reuse in prediction.
- Provide an interactive application that allows practical testing of the trained model.

System Workflow

End-to-End Machine Learning Workflow



Figure 1. End-to-end project workflow from NASA POWER acquisition to Streamlit deployment.

1. Introduction

Solar-energy planning depends on reliable estimates of the solar resource under changing location, season and weather conditions. Yemen has strong solar potential, but local decision-making benefits from a data-driven tool that can compare cities and estimate expected radiation from measurable inputs. This project implements a complete machine-learning lifecycle rather than presenting only a user interface.

1.1 Problem Statement

The core task is to estimate daily solar radiation (kWh/m²/day) using geographic, temporal and meteorological predictors. The target variable is Solar_Radiation_kWh_m2_day. Inputs include city, latitude, longitude, year, month, day, DayOfYear, cyclical month features, temperature, relative humidity and wind speed.

1.2 Scope

The dataset covers Aden, Hodeidah, Ibb, Marib, Mukalla, Sanaa, Seiyun and Taiz from 2015-01-01 through 2025-12-31. The trained model is intended for academic prediction and comparative analysis within the feature ranges represented by the dataset.

2. Data Acquisition

2.1 Source

Data are downloaded programmatically from NASA POWER (Prediction Of Worldwide Energy Resources) using the daily point API. The acquisition script requests one geographic point for each selected Yemeni city and writes a unified CSV dataset. This makes the data source reproducible and avoids manually assembled records.

2.2 Requested NASA POWER Parameters

API Parameter	Project Column	Meaning
ALLSKY_SFC_SW_DWN	Solar_Radiation_kWh_m2_day	All-sky surface shortwave downward solar radiation
T2M	Temperature_C	Air temperature at 2 m
RH2M	Relative_Humidity_pct	Relative humidity at 2 m
WS2M	Wind_Speed_2m_m_s	Wind speed at 2 m

2.3 Geographic Coverage

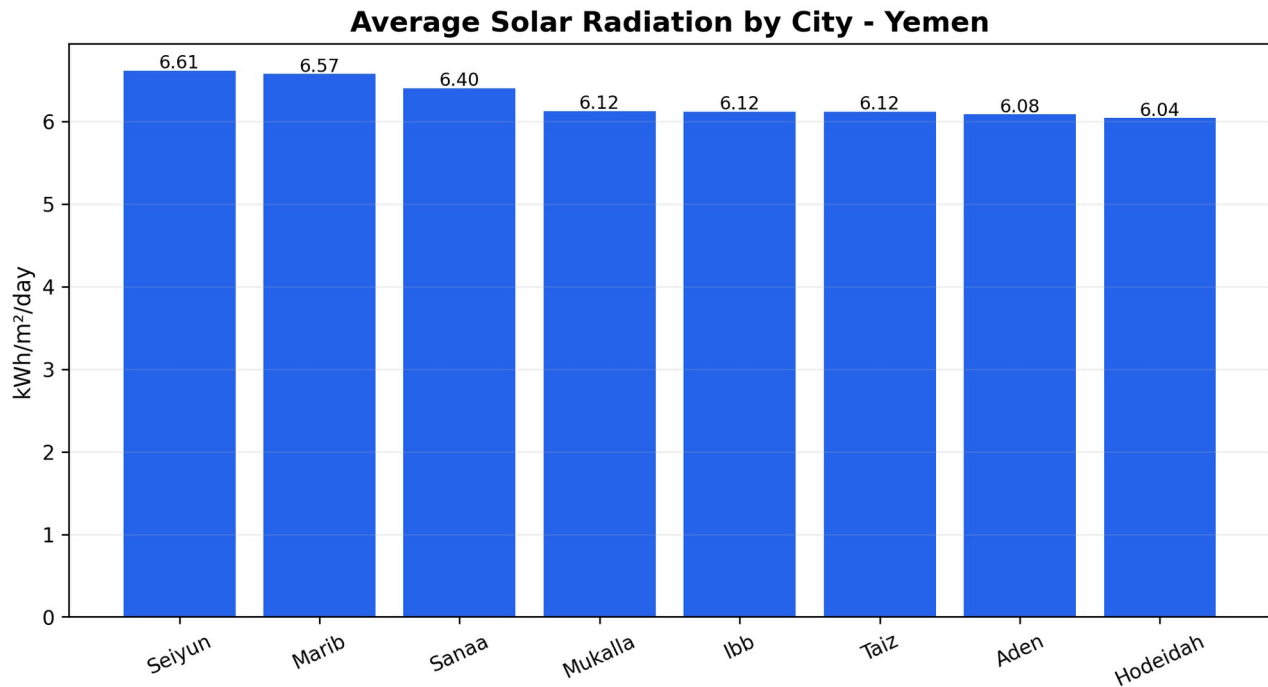


Figure 2. Mean solar radiation by city in the cleaned Yemen dataset.

3. Data Quality and Pre-processing

The cleaned dataset contains 32,144 records and 14 columns. The final training dataframe contains no missing values and no duplicated rows. Dates are parsed into calendar components and additional cyclical features are generated so that seasonality can be represented more naturally.

3.1 Feature Engineering

Feature	Purpose
Year, Month, Day	Calendar information
DayOfYear	Continuous position within the year
Month_Sin, Month_Cos	Cyclical encoding of monthly seasonality
City	Categorical geographic identity
Latitude, Longitude	Geographic position
Temperature, Humidity, Wind	Meteorological conditions

3.2 Train/Test Strategy

The dataset is divided into 80% training and 20% testing with `random_state=42`. City is one-hot encoded with `handle_unknown="ignore"`, while numeric predictors pass through unchanged. All candidate models use the same split and preprocessing pipeline, enabling a fair comparison.

4. Exploratory Data Analysis

Exploratory analysis is used to understand city-level differences, seasonal behavior and the relationship between weather variables and solar radiation before model interpretation. High-resolution plots are generated directly from the cleaned dataset rather than from phone screenshots.

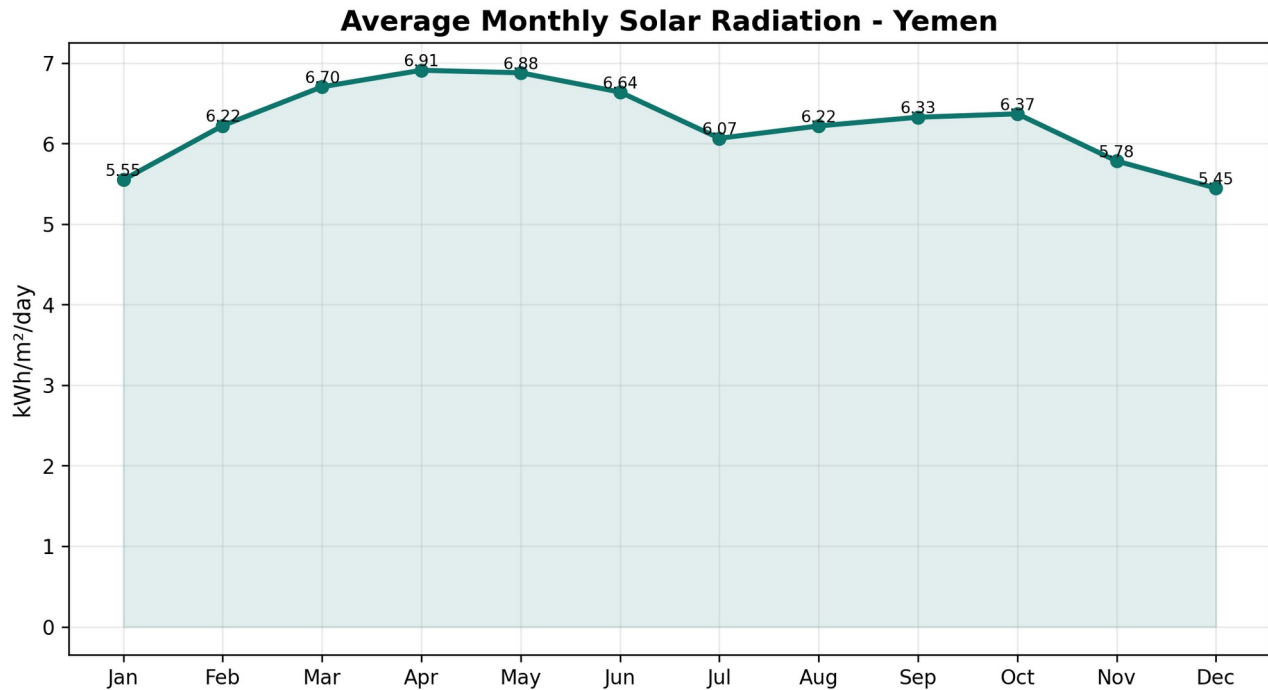


Figure 3. Average monthly solar-radiation profile across the eight project cities.

The monthly profile shows clear seasonality, with stronger average solar resource during spring and early summer and lower values toward the end of the year. This pattern motivates the use of Month, DayOfYear and cyclical month features in the model.

4.1 Correlation Analysis

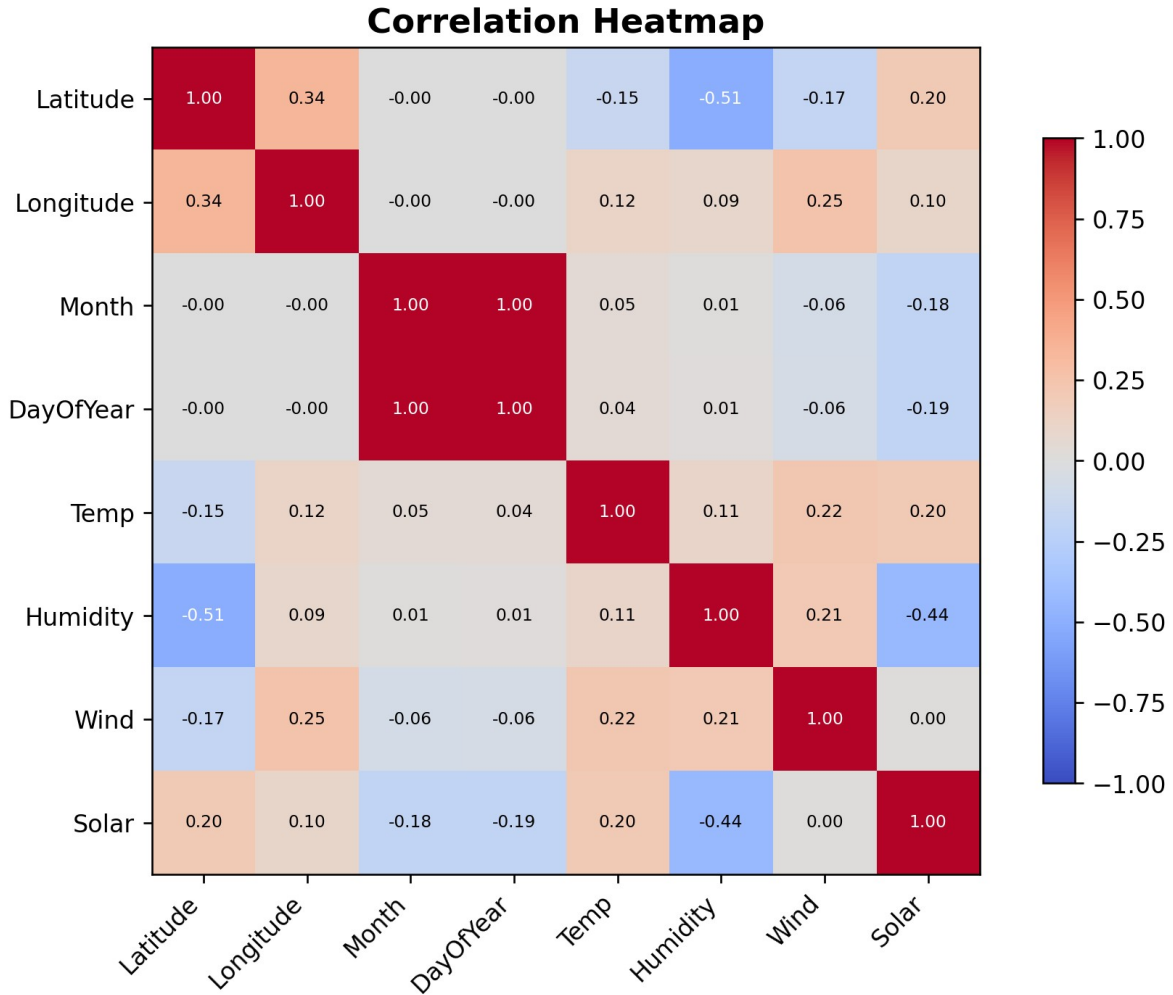


Figure 4. Correlation heatmap for selected numeric features.

Relative humidity has a noticeable negative association with solar radiation, while temperature shows a smaller positive relationship. Month and DayOfYear are naturally highly correlated because both represent seasonal position. Correlation alone does not determine importance, so nonlinear tree-based models are also evaluated.

5. Machine Learning Models

Three supervised regression algorithms are trained using the same preprocessing and train/test split so that their performance can be compared fairly.

Algorithm	Role	Project Setting
Linear Regression	Interpretable linear baseline	Default regression
Random Forest	Nonlinear ensemble of decision trees	150 estimators, random_state=42
Gradient Boosting	Sequential boosted-tree ensemble	150 estimators, random_state=42

5.1 Evaluation Metrics

- MAE - average absolute prediction error; lower is better.
- RMSE - penalizes larger errors more strongly; lower is better.
- R^2 - proportion of target variance explained by the model; higher is better.

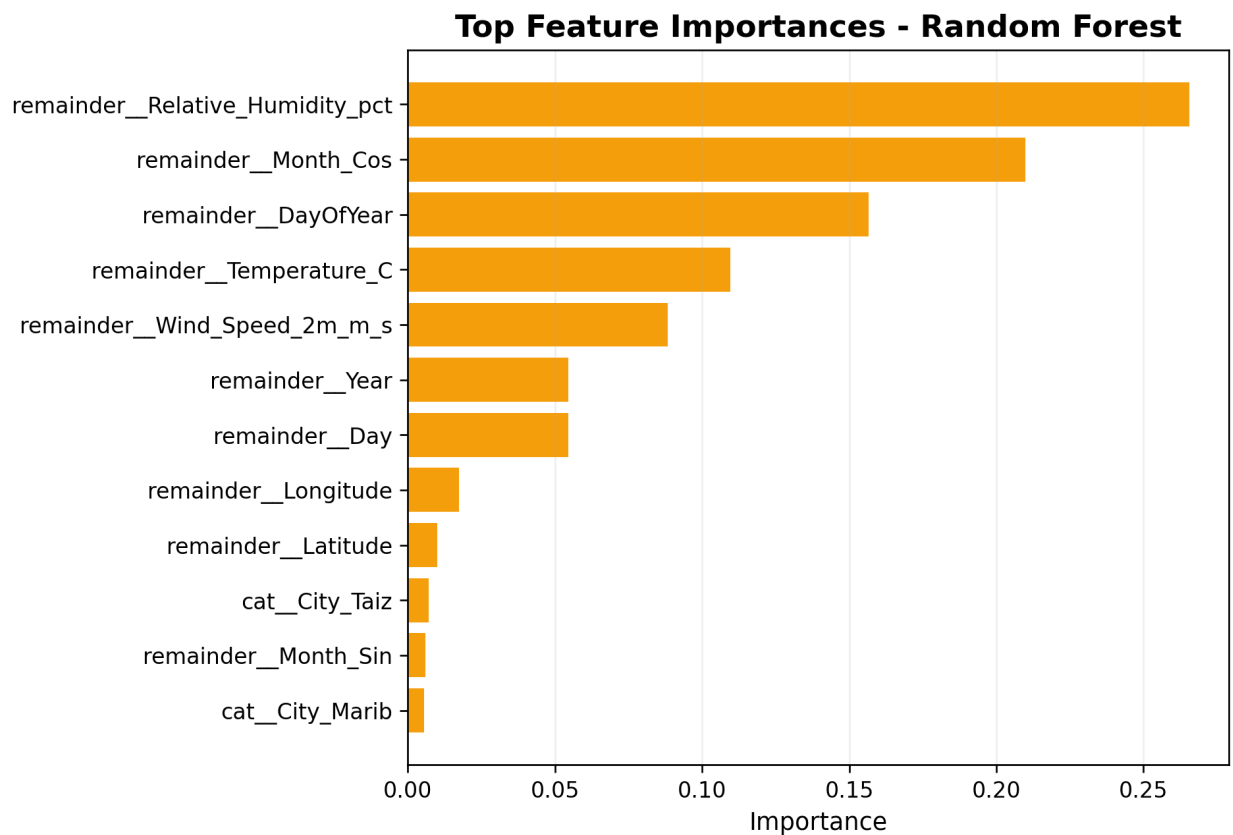


Figure 5. Top feature importances from the selected Random Forest pipeline.

6. Results and Model Comparison

Model	MAE	RMSE	R ²
Random Forest	0.3481	0.5054	0.6360
Gradient Boosting	0.3837	0.5432	0.5795
Linear Regression	0.4996	0.6569	0.3850

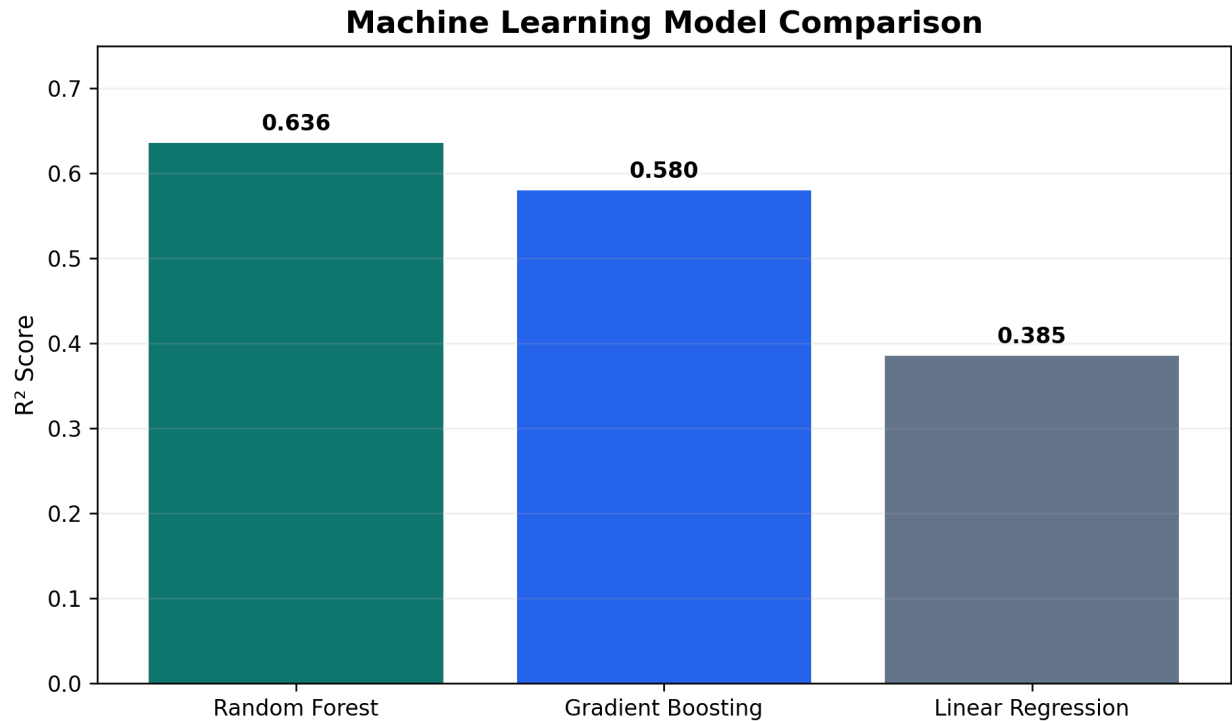
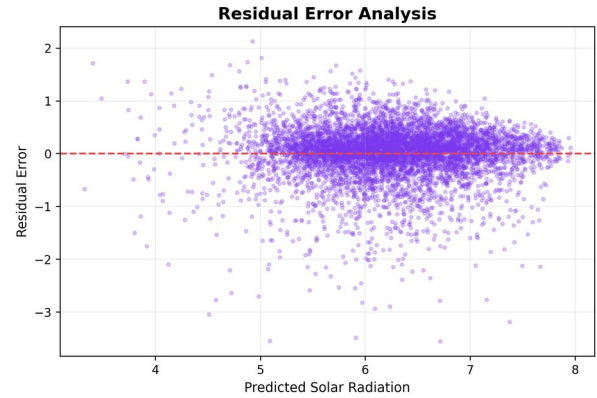
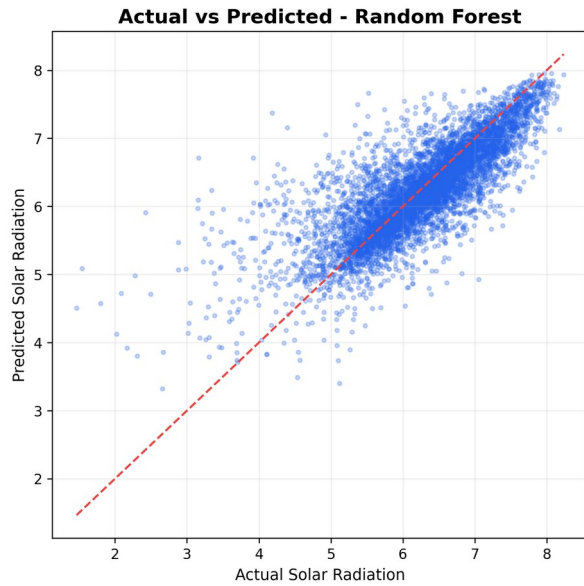


Figure 6. R² comparison for the three candidate regression algorithms.

Random Forest achieves the best overall test performance, with the highest R² and the lowest MAE/RMSE among the evaluated models. It is therefore selected and saved as the final prediction pipeline used by the application.

7. Model Diagnostics

Random Forest Diagnostic Views



The Actual-vs-Predicted plot shows predictions clustering around the ideal 45-degree line, while the residual view highlights remaining error spread. The model performs best in the dense central range of the dataset and shows larger deviations for less-common extreme observations.

8. Streamlit Application and Deployment

The original desktop interface has been upgraded to a Streamlit web dashboard for a cleaner and more professional user experience. The dashboard loads the saved model pipeline and cleaned dataset, then provides interactive controls for city, date, temperature, humidity, wind speed, system size and system efficiency.

- Executive dashboard with solar-radiation, suitability, daily-energy and annual-energy KPIs.
- Interactive prediction and 12-month model-based solar scenario forecast.
- EDA tab with city comparison, monthly profile, weather relationships and a correlation heatmap.
- Model Performance tab with MAE, RMSE, R^2 and feature importance.
- Interactive Yemen Solar Potential Map with city-level average radiation and suitability information.

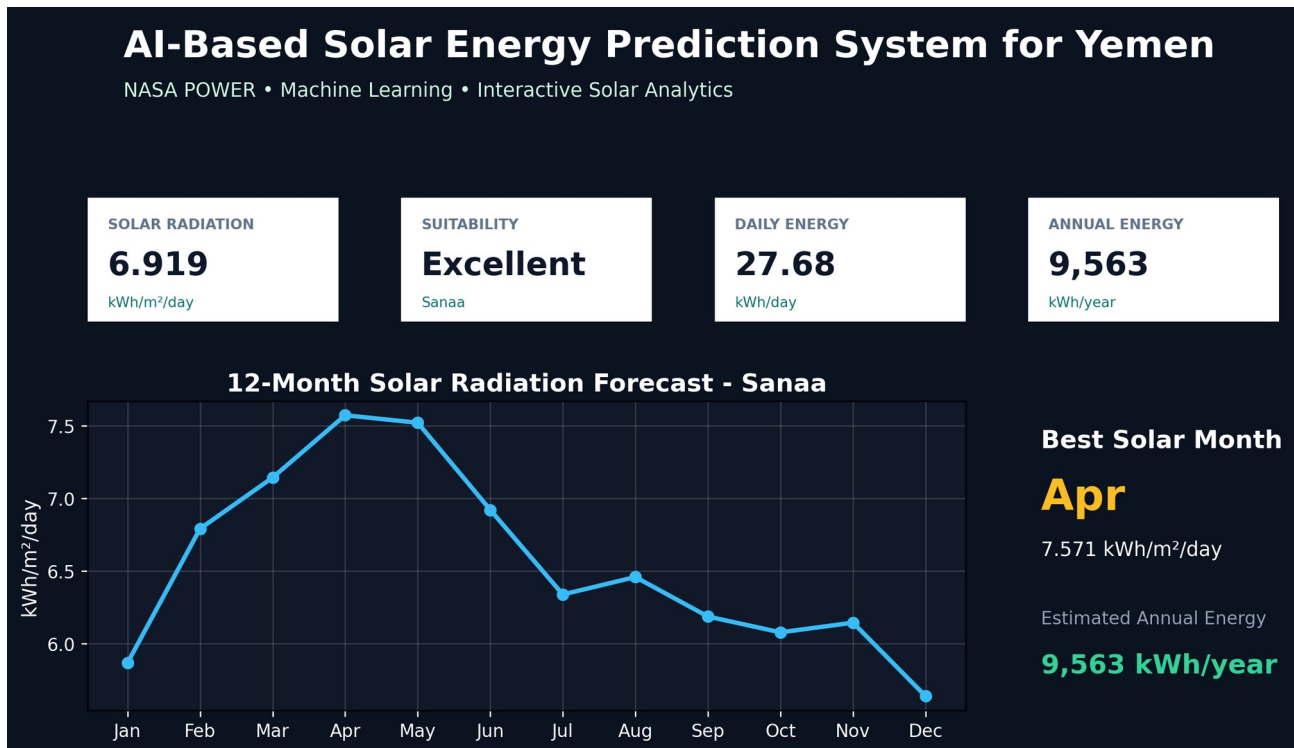


Figure 7. High-resolution representation of the final Streamlit dashboard; no phone screenshot is required in the report.

9. Compliance with Project Requirements

Requirement	Status	Evidence
Dataset from a reliable source	Complete	NASA POWER daily API
Data validation / preprocessing	Complete	Cleaned CSV, date parsing, engineered features
EDA and visualization	Complete	City, monthly, weather and correlation analysis
Multiple ML algorithms	Complete	Linear Regression, Random Forest, Gradient Boosting
Model evaluation	Complete	MAE, RMSE and R^2
Best-model selection	Complete	Random Forest selected automatically by R^2
Saved trained model	Complete	Reusable joblib pipeline
Practical test interface	Complete	Interactive Streamlit application
Geographic visualization	Complete	Yemen Solar Potential Map

Overall, the project satisfies the requested end-to-end machine-learning workflow: data acquisition, preprocessing, visualization, model training, algorithm comparison, performance reporting and an executable interface for model testing.

10. Conclusion and Recommendations

The project demonstrates a complete and reproducible machine-learning solution for solar-resource prediction in Yemen. Random Forest provides the strongest performance among the three tested algorithms and is deployed through an interactive Streamlit dashboard. The system goes beyond a simple predictor by combining model evaluation, exploratory analytics, energy estimation and geographic comparison in one application.

Recommendations

- Use location-specific weather forecasts when extending the system to real future forecasting.
- Add additional predictors such as cloud cover, pressure and aerosol indicators if available.
- Evaluate time-aware validation methods to better measure future-period generalization.
- Expand the geographic coverage to additional Yemeni districts and stations.
- Consider hyperparameter optimization and additional ensemble algorithms in a future version.

References

1. NASA POWER Project. Prediction Of Worldwide Energy Resources (POWER). <https://power.larc.nasa.gov/>
2. Scikit-learn documentation. Machine learning in Python. <https://scikit-learn.org/>
3. Streamlit documentation. <https://docs.streamlit.io/>
4. McKinney, W. Python for Data Analysis. O'Reilly Media.

Appendix A - Reproducibility Checklist

Step	Action	Expected Output
1	Run the NASA POWER download script	yemen_solar_nasa_power_2015_2025.csv
2	Run preprocessing / feature engineering	yemen_solar_cleaned.csv
3	Run EDA	High-resolution figures / interactive EDA
4	Run model training	model_comparison_results.csv + saved model
5	Start Streamlit application	streamlit_yemen_solar.py
6	Test at least one city and date	Prediction KPIs and forecast
7	Open Solar Potential Map	Interactive city markers and hover data

Key Project Files

- download_yemen_solar_data.py - downloads the NASA POWER dataset.
- preprocess_yemen_solar.py - cleans data and creates temporal features.
- eda_yemen_solar.py - generates exploratory figures.
- train_yemen_solar_models.py - trains, evaluates and saves the best model.
- streamlit_yemen_solar.py - final interactive dashboard.
- yemen_solar_cleaned.csv - final cleaned training dataset.
- model_comparison_results.csv - model performance summary.

This appendix provides the minimum sequence required to reproduce the final project demonstration on another machine with the required Python packages installed.